

Week 3 Discussion Sections

COGS 108 WI 2025

pandas and numpy

```
import pandas as pd
```

```
import numpy as np
```

pandas	numpy
When we have to work on Tabular data, we prefer the pandas module.	When we have to work on Numerical data, we prefer the NumPy module.
Pandas have a 2D table object called DataFrame.	Numpy is capable of providing multi-dimensional arrays.
The powerful tools of pandas are DataFrame and Series.	the powerful tool of NumPy is Arrays.
Pandas consume more memory.	Numpy is memory efficient.
Indexing of the Pandas series is very slow as compared to Numpy arrays.	Indexing of Numpy arrays is very fast.

Announcements

Project update:

- Should all be assigned to a Group and Github Repo
- Should have received an email with your Github Repo from Scott
- If you are not able to do any of the above PLEASE message STAFF to let us know **ASAP**



D3

- Feel free to get started on D3 if you haven't already and come up if you have any questions!
- We'll be running through some important points on it with you!
 - e.g. how your graphs are supposed to look like



I. Import all the necessary libraries

- `numpy ( np )`
- `pandas ( pd )`
- `matplotlib.pyplot ( plt )`
- `seaborn ( sns )`

```
[ ]: # YOUR CODE HERE  
raise NotImplementedError()
```

```
[ ]: assert np  
assert pd  
assert plt  
assert sns
```

What goes
HERE???

For the love of everything
you hold dear, please
don't leave this here.....

I. Import all the necessary libraries

- `numpy ( np )`
- `pandas ( pd )`
- `matplotlib.pyplot ( plt )`
- `seaborn ( sns )`

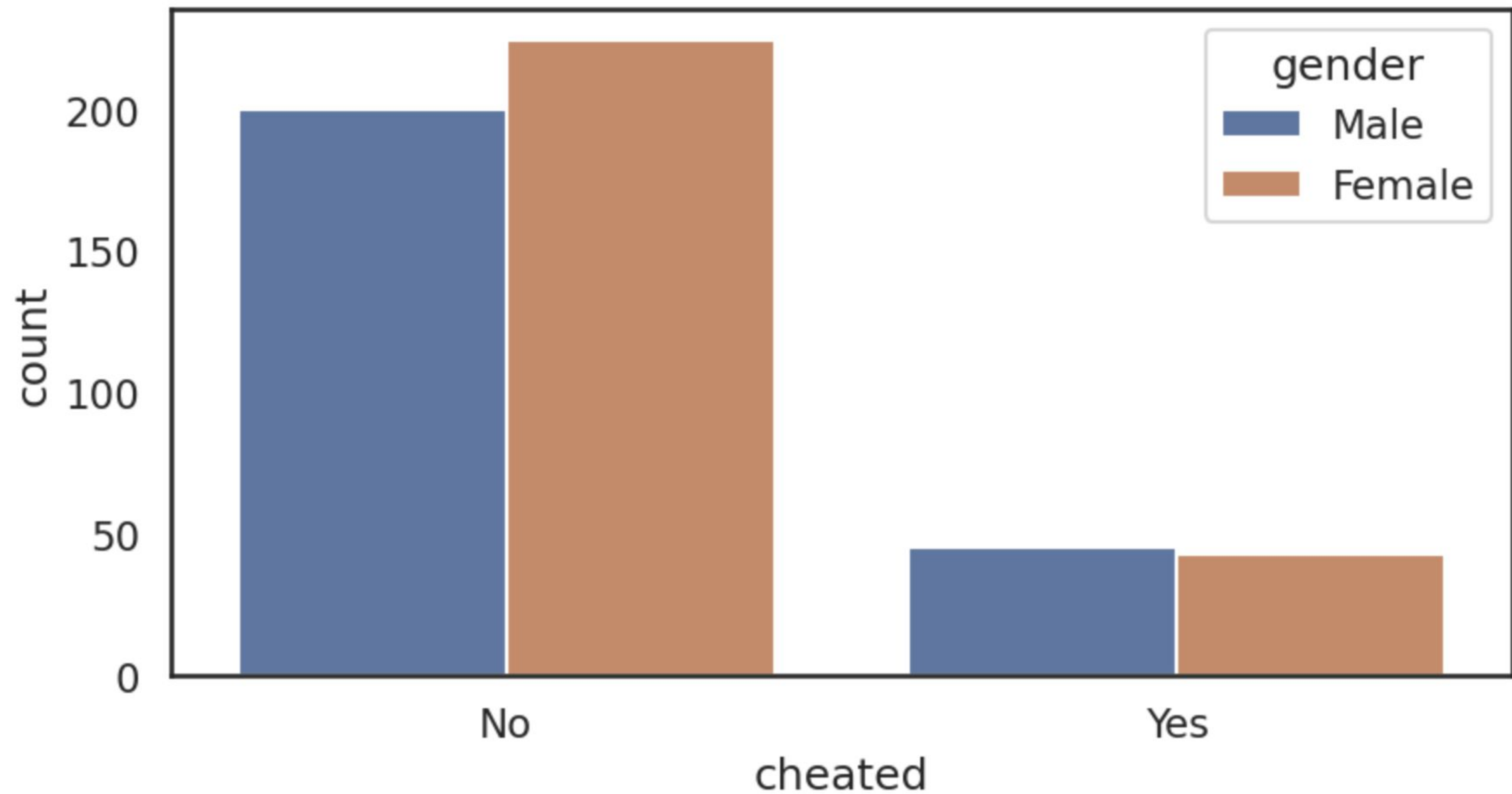
```
[1]: # YOUR CODE HERE
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
[2]: assert np
      assert pd
      assert plt
      assert sns
```

GOT IT?



2. PART I: CHEATING (plot_cheated)



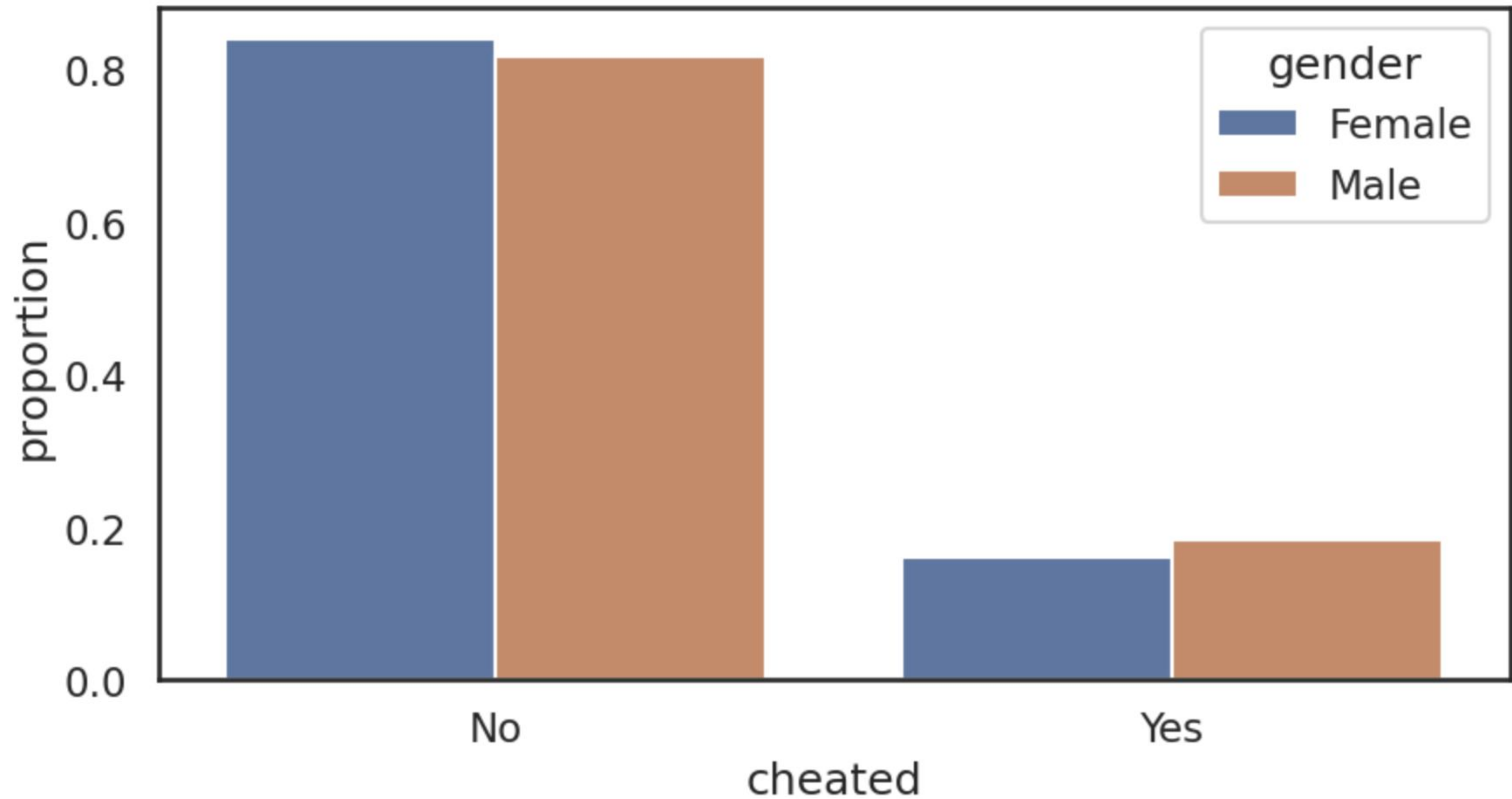
2. PART 1: CHEATING (prop_df)

	gender	cheated	proportion
0	Female	No	0.84
1	Female	Yes	0.16
2	Male	No	0.82
3	Male	Yes	0.18

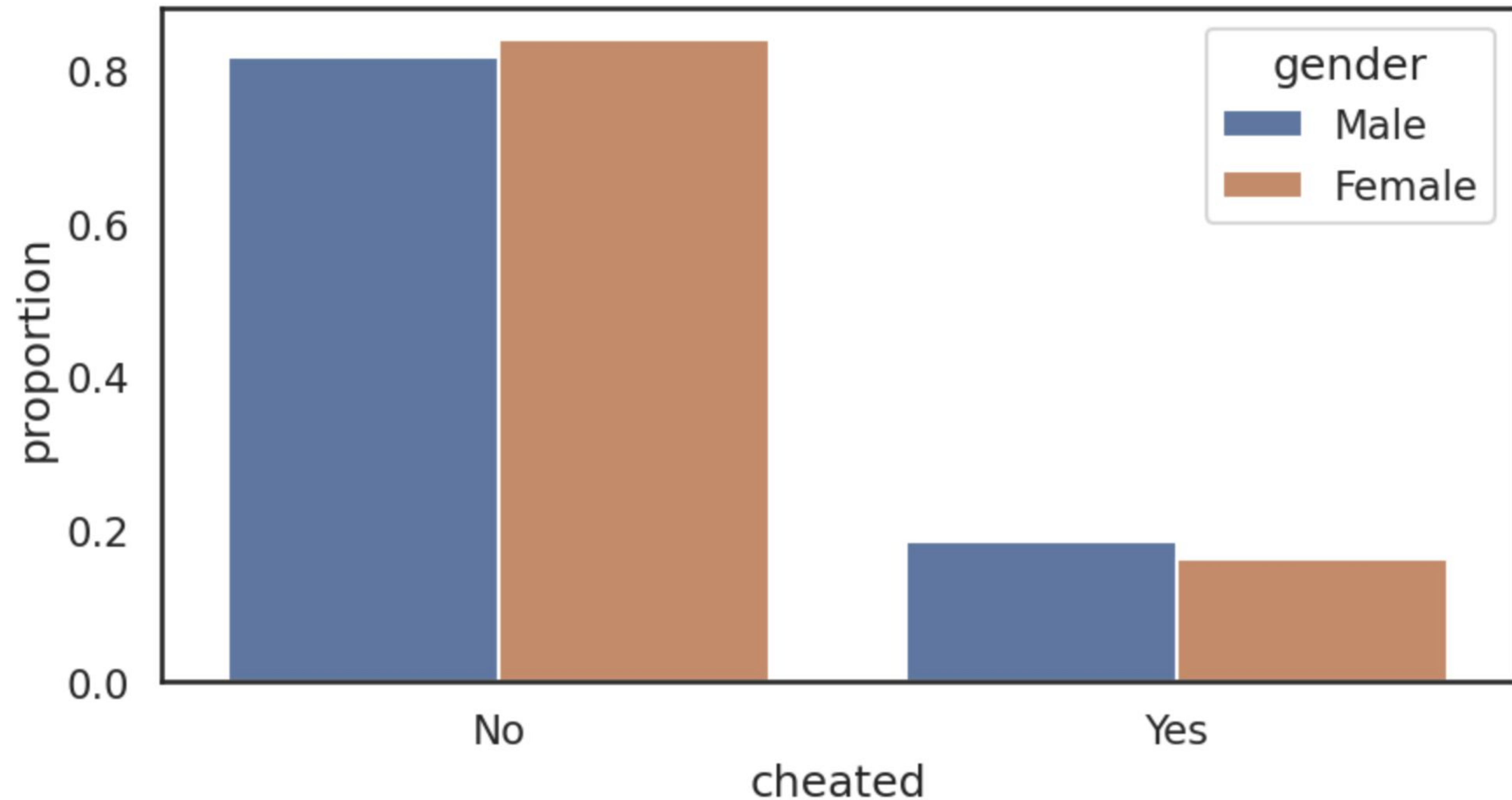
I need to make
this data into a
pretty and
easy to read
visualization?!?



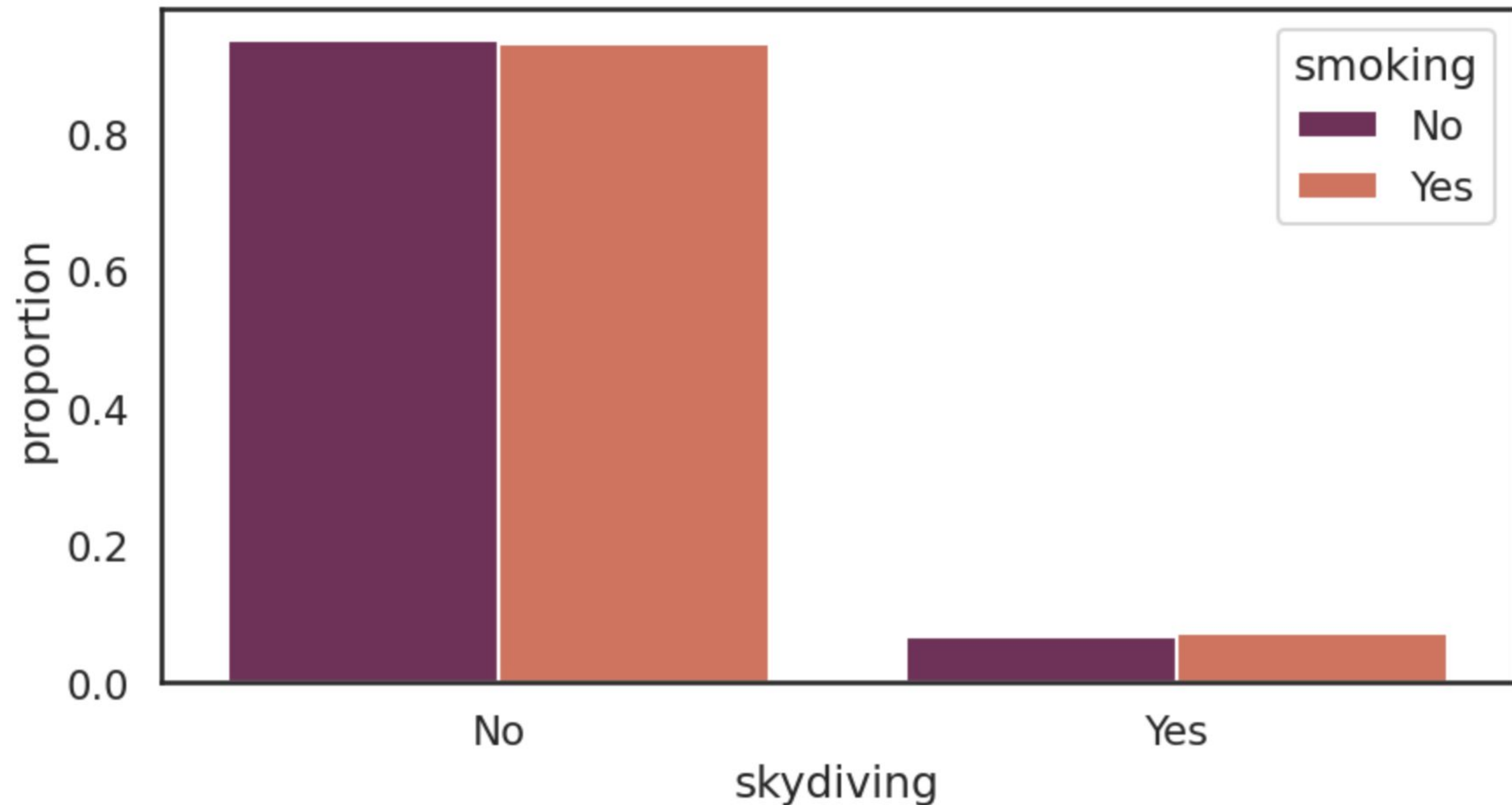
2. PART I: CHEATING (plot_proportion)



2. PART I: CHEATING (plot_reorder)



3. PART II: RISKY BEHAVIOR (plot_skydiving)



4. PART III: GAMBLING (plot_aamblina)

